

Home AI — Platform PRD

Product Requirements Document v1.1 — July 2026

Working title: "Home AI" (trademark check pending). Built from the Silky Oak app project.

1. Product summary

A white-label, AI-native home platform sold to builders and developers, used by homeowners and residents. Each property gets a living digital twin — plans, appliances, finishes, warranties, key dates, and documents — with a conversational AI that knows the property, a structured defect and maintenance system, and a branded portal for the organisation that provisioned it.

Positioning: competitors hand over documents; we hand over answers. HouseLab, Constructive, and ClickHome stop at document logistics. This product is the assistant layer on top — and starts where they stop: at handover.

2. Problem statement

Builders and developers spend \$500–\$2,000 per project on handover and post-occupancy admin, then bleed time on repetitive homeowner calls and unstructured defect claims. Homeowners inherit a folder of PDFs they never read and a house they don't understand. No product in the Australian market gives the property itself a brain that both parties can talk to.

3. Goals

1. A builder can provision a fully-loaded home in **under 15 minutes** (template clone +

doc sideload); a 20-unit multi-res building in **under 90 minutes**.

2. **≥40% monthly active homeowners** across pilot portfolio at 90 days post-handover.
3. **≥30% of would-be defect/support contacts deflected** by AI triage (see Defect Module Spec v1.0).
4. Platform AI cost **<\$1.50/property/month** on the included tier; zero marginal AI cost on BYO-key organisations.
5. Two paying organisations (one builder, one multi-res developer) within 6 months of pilot start.

4. Non-goals (v1)

- **No build-phase features** (progress tracking, selections, scheduling) — Constructive/ClickHome territory; we begin at handover.
- **No native iOS/Android apps** — PWA on mobile and desktop. Revisit if enterprise deals demand it.
- **No trade accounts** — magic-link work orders only (per Defect Module Spec).
- **No payments/marketplace** (servicing packages) — data model allows it; not built.
- **No body-corporate/strata management** (levies, meetings, by-law workflows) — MYBOS/BuildingLink territory. We hold building-level *documents and defects*, not strata governance.
- **No short-stay / guest ("Airbnb") mode** — descoped entirely. The tenancy model does not preclude a future guest role; nothing is designed or built for it.

5. Personas

Persona	Context	Primary jobs
Org admin — builder ("Bec")	Client care at 60-home/yr builder	Provision homes, sideload docs, manage defects, watch portfolio dashboard

Persona	Context	Primary jobs
Org admin — developer (multi-res) ("David")	Delivery manager, 40-unit project	Bulk-provision units, manage building-level assets + defects, hand to building manager
Homeowner ("Hannah")	New build, 4 months in	Ask the house questions, report issues, track maintenance/warranties, add her own docs
Resident (multi-res)	Apartment owner/occupier	Same as homeowner + building-level info (bins, amenities, building manager contact)
Platform admin (Mark)	Operator	Tenant management, AI usage/cost, deflection metrics, billing

6. Tenancy & permissions model

`Organisation (builder | developer)
└ Property group (optional: estate / building / portfolio)
 └ Property (home | unit)
 └ Users: owner, member, resident (+ time-limited guest codes for
 access sharing, carried over from Silky Oak)`

Permissions matrix (summary)

Capability	Org admin	Owner	Member	Resident
Provision/edit property structure	✓	-	-	-
Sideloading org docs (read-only to owner)	✓	view	view	view*
Upload/edit own docs	-	✓	✓	✓ (unit)
AI chat (full property context)	✓ (per property)	✓	✓	✓
Report defects	-	✓	✓	✓
Assess/assign defects	✓	-	-	-
Building-level docs/defects (multi-res)	✓	-	-	view / common-area reports
Share guest access codes	-	✓	-	-

* Residents see building-level docs flagged "resident-visible."

Document layers (core design rule):

1. **Org layer** — sideloaded at setup, read-only to the homeowner, snapshotted as an immutable **archive copy** at handover (dispute evidence — the HouseLab pattern).
2. **Owner layer** — homeowner's own uploads/edits (receipts, renovation docs, photos). Fully editable by them, portable if they leave the platform.
3. **Shared asset layer** — org-level library inherited by properties (see §8).

The AI reads all layers the current user is entitled to, and only those.

7. Feature areas

7.1 Developer portal (web, desktop-first)

P0

- Org onboarding: branding (logo, colours, hero image), ABN, team seats with roles.
- Property templates: define a design/spec once (appliance set, finishes schedule, standard docs); clone per property; edit deltas only.
- Bulk provisioning: CSV or table entry of addresses/lots → template assignment → batch create. Multi-res: building shell + unit generation (unit numbering patterns).
- Doc sideload: drag-and-drop folders with **tag-preserving ingestion** (folder path → tags) + AI extraction (model numbers, warranty periods, serials → structured appliances records). Review screen before commit.
- Handover action: assign owner email → invitation → archive copy locked.
- Defect management screens per Defect Module Spec v1.0 (queue, detail, assign, portfolio dashboard).
- Portfolio dashboard: activation, MAU, defects by status/trade/product, deflection rate.

P1

- Announcements/content push to owners (org-branded, AI-drafted option).
- NPS pulse configuration (3/9-month post-handover + defect closure).
- Ledger PDF exports (per property, per building).

P2

- Insight reports across portfolio (product failure rates, trade performance benchmarks).

- API/webhooks for builders' own systems.

7.2 Homeowner app (PWA, mobile-first; same app responsive on desktop)

P0

- Auth: email magic link + OAuth (Google/Apple). Household members invited by owner. (PIN gate retired.)
- Four-tab structure carried over: **Chat, Dates, House, Issues** (Deals becomes P1, see below).
- Chat: conversational AI with per-property RAG context (see §7.4), nudges strip, Snap & Diagnose, voice input, file/camera upload.
- Dates: bins (council-schedule aware), auto-rescheduling tasks, warranty countdowns + claim-draft helper, crack log, per-event calendar download.
- House: emergency shut-offs, plans, finishes board, appliance registry (linked doc bundles + maintenance ledger per appliance), service directory (org-preloaded trades + owner's own), guest codes/share cards.
- Issues: defect intake per Defect Module Spec (triage → deflect/escalate/emergency), status tracking, sign-off.
- Owner doc upload/edit: add documents, attach to rooms/appliances, edit their own layer; org layer read-only with "request correction" flow.
- Share-spec cards: native share of a paint colour, tile spec, or plan extract to a tradie (Chimni-validated need).

P1

- Deals: renewal watching + AI market scan (needs legal review re: financial-product comparison framing before commercial release — reposition as "renewal reminders + market information").
- Maintenance ledger surfaced against warranty terms ("proof of servicing" pack).

- NPS pulses.

P2

- Resale transfer: hand the property record to the next owner (agent channel play).
- Freemium self-serve homeowner tier (homes not provisioned by an org).

7.3 Multi-res mode

A building is a property group with its own document set, appliance/plant registry (lifts, pumps, fire systems — visible to org/building manager only), and defect queue for common areas. Units are properties inheriting building-level docs flagged resident-visible (waste schedules, amenity guides, building manager contact).

P0: building shell + unit generation; building-level doc layer; resident role; common-area defect reporting routed to building manager (an org seat). **P1:** building announcements; building-level ledger export (Shergold Weir digital building manual alignment — sales angle: the platform is the comprehensive building manual with an AI on top). **P2:** strata handoff package (export for owners corp at settlement of final unit).

7.4 AI layer

Included AI (default): platform-provided Claude access, per-property rate limits (config per plan tier), standard model for chat/triage, premium model reserved for Snap & Diagnose and (P1) Deals scans. Hard monthly token budget per org with soft-warning at 80%; over-budget degrades gracefully to "AI paused — resumes on the 1st or upgrade" rather than erroring.

BYO AI (org-level setting): org supplies their own Anthropic API key (stored encrypted, server-side only, per-org isolation; never exposed to clients). All AI calls for that org's properties route via their key → their cost, our margin protected, no rate caps beyond abuse limits. Validation ping on save; automatic fallback to *paused* (not to platform key) if their key fails, with admin alert.

- v1 supports Anthropic keys only. Provider abstraction in the gateway so OpenAI-compatible endpoints can be added later (P2) — do not promise multi-provider at launch.

Context architecture: per-property structured profile (from ingestion) injected as system context + RAG over the property's document layers scoped to the requesting user's entitlements. All prompts/responses logged per tenant for cost, quality, and dispute audit.

Guardrails (global, non-overridable by BYO orgs): emergency hard-rules pre-LLM; no liability determinations; licensed-professional framing for gas/electrical/structural; per-session and per-day message caps against abuse.

7.5 Shared assets

Org-level asset library: an appliance/product/document defined once (e.g. "Bosch SMS6HCI01A — manual, warranty terms, quick-start") and inherited by every property that references it. Template cloning pulls from the library. Update propagation: library edits flow to linked properties with change-log entry; org can pin a property to a version.

P0: library CRUD, inheritance, template integration. **P1:** usage counts ("this dishwasher is in 43 of your homes"), defect-rate-by-product feed into portfolio dashboard. **P2:** cross-org anonymised product benchmarks (opt-in) — a future data product.

8. Platform & technical requirements

- **Stack:** React front-ends (homeowner PWA + portal), Supabase (Postgres, RLS, Storage, Auth), Vercel hosting, serverless AI gateway. Aligns with existing PP stack and skills.
- **Multi-tenancy:** RLS on every table keyed by org/property; **data-isolation audit is a launch gate** for each pilot cohort.
- **Ingestion pipeline:** upload → tag-preserving structure capture → AI extraction pass → human review screen → commit to structured records + vector index. Target: 50

docs processed in <10 minutes wall-clock.

- **Environments & deploy:** GitHub + CI/CD, staging + production, Sentry, uptime monitoring. The copy-paste deploy loop is retired before any external user touches the product.
- **Compliance:** Australian Privacy Act privacy policy; AU region hosting where available; data export + deletion per owner; ToS + AI disclaimers (solicitor-reviewed).
- **Billing:** Stripe; per-property annual SaaS + one-off provisioning fee; plan tiers gate AI limits and seats.

9. Success metrics

Leading (first 90 days): provisioning time per property; % properties activated (owner accepted invite) within 14 days of handover (target $\geq 70\%$); homeowner WAU/MAU; triage deflection rate $\geq 30\%$; AI cost/property/month $< \$1.50$.

Lagging (6–12 months): builder-reported support-contact reduction; defect median resolution < 21 days; NPS (homeowner ≥ 40 , org admin ≥ 50); pilot $\rightarrow$ paid conversion; logo count (2 orgs by month 6); churn = 0 in year 1 cohort.

10. Phasing

- **Phase 0 (wks 1–6):** CI/CD, multi-tenant schema, auth, ingestion v1, migrate 18 Silky Oak as tenant #1. Defect Sprint 1 (per module spec).
- **Phase 1 — Builder pilot (wks 6–16):** portal P0, homeowner app P0, defect Sprints 2–3, shared assets P0. One builder, 20–50 homes, concierge ingestion. Isolation audit gate.
- **Phase 2 — Second vertical (months 4–6):** multi-res P0 with a developer pilot; BYO AI shipped for the first org that asks (it's small — gateway keying — build on demand); notification layer hardening (email/push per event type).

- **Phase 3 (months 6+):** P1 backlog, portfolio insights, agent/resale channel, freemium homeowner tier.

11. Open questions

- **Blocking / legal:** AI disclaimer + deflection duty-of-care wording (shared with Defect Module Spec); Deals feature vs financial-comparison regulation.
- **Blocking / commercial:** product name + trademark; pricing tiers finalised before pilot contracts (working model: \$150–300 provisioning + \$5–10/property/month org-billed).
- **Non-blocking / product:** does the org admin see homeowner chat transcripts? (Recommendation: **no** — aggregate topics only; privacy is a selling point to homeowners and a risk shield for orgs.) Confirm.
- **Non-blocking / engineering:** vector store choice (pgvector in Supabase vs external); EXIF/GPS stripping policy on media.

12. Assumptions made in this PRD (flag if wrong)

1. **BYO AI = org-level Anthropic key**, not homeowner-level. Homeowners always use the org's allocation.
2. **"Desktop version" = the same responsive PWA on desktop**, plus the portal which is desktop-first — no separate desktop client.
3. **Org docs are read-only to homeowners** (with a request-correction flow); homeowner-editable docs live in their own layer. Keeps the archive copy defensible.
4. **Multi-res excludes strata governance** — documents, assets, and defects only.

